PHASE 1: STAGE 1 PREDICTIVE CORE (OFFLINE TRAINING)

Training Split D_train (80%)

Standardized Stratified Split
PaySim / IEEE-CIS / ULB

Stage 1: XGBoost Core

Cost-Weighted Tree Ensemble f_core(x)
Regularized Logistic Loss (spw=10.0)

Deploy f_core

PHASE 2: STAGE 2 VALIDATION & AMOUNT-STRATIFIED CALIBRATION

Validation Split D_val (10%)

Score Probabilities: p_val = f_core(x)
Standardized Held-Out Partition

Stage 2: Cost-Optimal Tuning (τ*)

100-step Grid Search τ ∈ [0.01, 0.99]
PaySim: τ* = 0.2179 | IEEE-CIS: τ* = 0.3763

Candidate Alerts (p >= τ*)

k=5 Stratified OOF Cross-Fitting

1. Extract Candidate Pool D_cand (p >= τ*)
2. Exact Local TreeSHAP Attributions ϕ(x)
3. Cross-Fit Random Forest Meta-Classifier

OOF Risk Scores q

Amount-Stratified Safety Calibration

Stratify Alerts into K=3 Amount Tertiles
Calibrate Tier Guards: τ_meta,k (ΔFN = 0)
Zero Additional False Negatives Observed

PHASE 3: STAGE 3 PRODUCTION STREAMING INFERENCE WITH SAFETY GUARD

Incoming Transaction

Streaming Vector x ∈ R^d
Fast Feature Extraction
Latency: 0.002 ms

Stage 1 Fast-Path

Evaluate f_core(x) -> p
Latency: ~0.55 ms
Decision Gate: p >= τ* ?

p < τ*

Auto-Cleared Legit

Direct Approval (Bulk Benign Traffic)

Alert: p >= τ*

Stage 3 TreeSHAP

Local Attribution ϕ(x)
Exact Tree Explainer
Latency: 3.49 ms

Meta-Classifier

Joint Space [x, ϕ(x)]
RF Ensemble Inference
Output: Meta-Risk q (4.95 ms)

Safety Guard Gate

Tier k(Amount)
Evaluate:
q >= τ_meta,k ?

q >= τ_k

Suppressed Alarm

Auto-Cleared FP (s = 1)
Zero Added FNs

q < τ_k

Escalated Review

Confirmed Fraud (s = 0)
Priority Compliance Queue